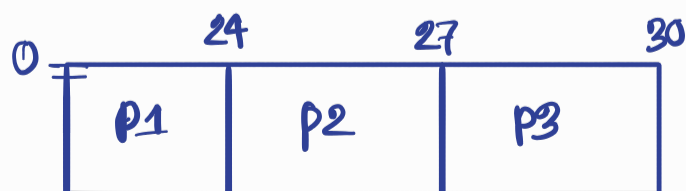


Process $\rightarrow$ I/O + CPU

Farhan @ first

FCFS Algorithm:
~~~~~

|                | <u>Burst-time</u> | <u>Arrival-time</u> |
|----------------|-------------------|---------------------|
| P <sub>1</sub> | 24                | 0                   |
| P <sub>2</sub> | 3                 | <u>1</u>            |
| P <sub>3</sub> | 3                 | 2                   |

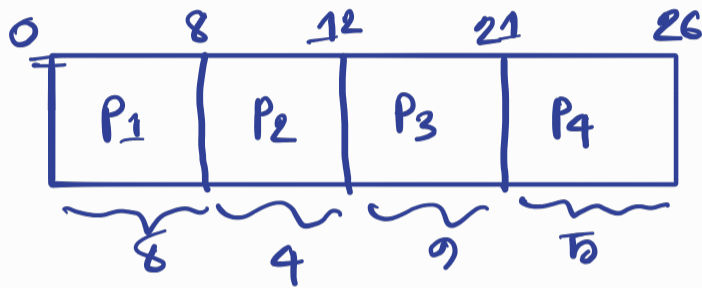


Gantt chart

$$P_1(\text{waiting}) : 0$$

$$P_2(\text{waiting}) : (24-1)$$

$$P_3(\text{waiting}) : (27-2)$$



$$P_1(\text{waiting}) : 0$$

$$P_2(\text{waiting}) : (8-1)$$

$$P_3(\text{waiting}) : (12-2)$$

$$P_4(\text{waiting}) : (21-3)$$

---


$$\text{Average (waiting)} = \frac{0+7+10+16}{3}$$

#SJF:  
mmr

Anive or not  $\rightarrow$  choose smallest burst time

|                | Bt | At |
|----------------|----|----|
| P <sub>1</sub> | 24 | 1  |
| P <sub>2</sub> | 3  | 2  |
| P <sub>3</sub> | 4  | 0  |



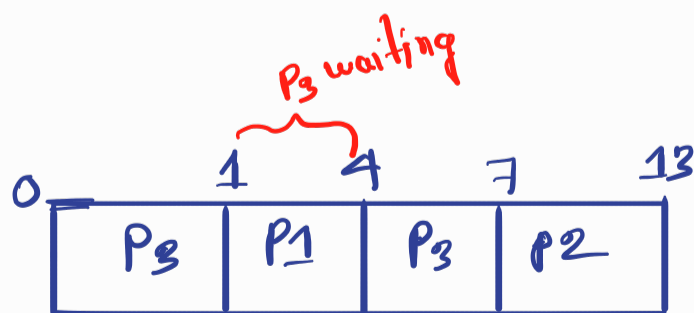
2 type Process:

primitive  $\longrightarrow$  Interruptable

non-primitive  $\longrightarrow$  non interruptable

# SRTF: સર્વે CPU બાદમાં Direct Lowest B.T (SJF)

|                | Bt              | At |
|----------------|-----------------|----|
| P <sub>1</sub> | <del>3</del> 20 | 1  |
| P <sub>2</sub> | 6               | 2  |
| P <sub>3</sub> | <del>4</del> 3  | 0  |



P<sub>1</sub> (waiting) : 0

P<sub>2</sub> ( ) : (7-2)

P<sub>3</sub> ( ) : 0 + (4-1)

|       |       |       |       |       |       |       |       |    |
|-------|-------|-------|-------|-------|-------|-------|-------|----|
| 0     | 1     | 2     | 3     | 4     | 7     | 11    | 18    | 26 |
| $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_2$ | $P_4$ | $P_1$ | $P_3$ |    |

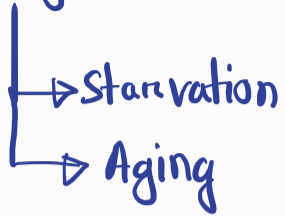
~~8~~ 7

~~4~~ ~~5~~ 6

~~9~~ 8

~~10~~ ~~4~~

Priority



RR



when T (def) RR  $\rightarrow$  FCFS

MLFQ  $\rightarrow$  Feedback Queue.

Time

Context switch  $\uparrow \rightarrow$  Store/Load  $\uparrow \rightarrow$  Overhead  $\uparrow$